Augmented ESKF Mathematical Formulation

sensor-fusion documentation

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1 Scope

This document describes the runtime augmented ESKF implemented by `ekf/src/rust_eskf.rs`, `ekf/src/generated_eskf.rs`, and the included generated model files. The filter estimates NED attitude, velocity, position, IMU biases, and a residual seeded-frame-to-vehicle mount quaternion. It is the primary runtime inertial/GNSS filter behind `SensorFusion`.

2 Frames and Mount Sources

Symbol	Meaning
n	Local NED frame. ESKF position and velocity states are in this frame.
b	Raw IMU body frame.
v	Physical vehicle frame, FRD: forward, right, down.
s	Seeded vehicle frame after raw IMU samples are pre-rotated by a coarse mount seed.
c	Corrected vehicle frame after applying the residual mount quaternion q_{cs}.

Rotations follow

$$x_a = C_{ab} x_b, \qquad C(q_1 \otimes q_2) = C(q_1)C(q_2). \tag{1}$$

The align filter supplies a physical vehicle-to-body mount q^0_{vb}. The fusion facade pre-rotates each raw IMU delta into the seeded frame:

$$\Delta\theta_s = C^0_{sv} C^0_{vb} \Delta\theta_b = C^{0^T}_{bv} \Delta\theta_b, \qquad \Delta v_s = C^{0^T}_{bv} \Delta v_b. \tag{2}$$

If `EskfMountSource::LatchedAlign` is selected, q^0_{vb} is frozen at ESKF initialization. If `FollowAlign` is selected, the pre-rotation source follows the current align mount and the residual mount is reset to identity. A fixed external mount is also supported.

3 Nominal and Error States

The nominal state has 20 scalar components:

$$x = \begin{bmatrix} q_{ns} & v_n^T & p_n^T & b_g^T & b_a^T & q_{cs} \end{bmatrix}^T, \tag{3}$$

where q_{ns} rotates seeded-frame vectors into NED, v_n is NED velocity, p_n is local NED position, b_g and b_a are additive gyro and accelerometer biases in the seeded frame, and q_{cs} rotates seeded-frame velocity into the corrected vehicle frame.

The 18-dimensional error state is

$$\delta x = \begin{bmatrix} \delta\theta_s & \delta v_n & \delta p_n & \delta b_g & \delta b_a & \delta\psi_{cs} \end{bmatrix}^T. \tag{4}$$

The Rust index order is:

Indices	Error component
0..2	attitude small angle $\delta\theta_s$
3..5	velocity error δv_n
6..8	position error δp_n
9..11	gyro-bias error δb_g
12..14	accelerometer-bias error δb_a
15..17	residual mount error $\delta\psi_{cs}$

4 Nominal Propagation

The ESKF consumes one-sample IMU increments already expressed in the seeded frame:

$$u_k = \{\Delta\theta_s, \Delta v_s, \Delta t\}. \tag{5}$$

Bias-corrected increments are

$$\Delta\theta_s^c = \Delta\theta_s - b_g\Delta t, \qquad \Delta v_s^c = \Delta v_s - b_a\Delta t. \tag{6}$$

The generated nominal equations implement

$$q_{ns}^+ = \text{normalize}\left(q_{ns}^- \otimes \delta q(\Delta\theta_s^c)\right), \tag{7}$$
$$v_n^+ = v_n^- + C_{ns}(q_{ns}^+)\Delta v_s^c + g_n\Delta t, \tag{8}$$
$$p_n^+ = p_n^- + v_n^+\Delta t, \tag{9}$$
$$b_g^+ = b_g^-, \qquad b_a^+ = b_a^-, \qquad q_{cs}^+ = q_{cs}^-, \tag{10}$$

with $g_n = [0, 0, 9.80665]^T$ in NED.

5 Error-State Propagation

The covariance prediction uses generated discrete matrices:

$$\delta x_{k+1} = F_k\delta x_k + G_k w_k, \qquad P_{k+1}^- = F_k P_k^+ F_k^T + G_k Q_k G_k^T. \tag{11}$$

The 15-noise vector is

$$w = \begin{bmatrix} n_g & n_a & n_{bg} & n_{ba} & n_m \end{bmatrix}^T. \tag{12}$$

The diagonal covariance entries assembled at runtime are

$$Q_k = \text{diag}\left(\sigma_g^2\Delta t^2 I_3,\, \sigma_a^2\Delta t^2 I_3,\, \sigma_{bg}^2\Delta t I_3,\, \sigma_{ba}^2\Delta t I_3,\, \sigma_m^2\Delta t I_3\right). \tag{13}$$

If residual mount states are frozen, the mount random-walk block is set to zero and the mount covariance rows/columns are cleared.

6 Measurement Updates

Scalar observations use

$$r = z - h(\hat{x}), \tag{14}$$
$$S = HPH^T + R, \tag{15}$$
$$K = PH^T S^{-1}, \tag{16}$$
$$\delta x = Kr. \tag{17}$$

Covariance is updated with a scalar Joseph-form expression. The lateral and vertical NHC rows may be fused as one sequential two-row batch so the second row sees the covariance and accumulated correction from the first row.

6.1 GNSS Position and Velocity

GNSS position and velocity are local NED:

$$h_p(x) = p_n, \qquad H_p = \begin{bmatrix} 0 & 0 & I_3 & 0 & 0 & 0 \end{bmatrix}, \tag{18}$$

$$h_v(x) = v_n, \qquad H_v = \begin{bmatrix} 0 & I_3 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{19}$$

The generated GNSS rows have no direct residual-mount Jacobian. The runtime has optional mount update scales for GNSS rows; with the current generated H these only affect mount updates induced through covariance cross-correlation. The default public `fuse_gps` path passes zero scales.

6.2 Zero Velocity

The zero-velocity pseudo-measurement is a GNSS-velocity-shaped update:

$$z_v = 0, \qquad h_v(x) = v_n. \tag{20}$$

It does not directly observe residual mount because it is expressed in NED, not in corrected vehicle coordinates.

6.3 Vehicle Speed and NHC

Predicted seeded-frame velocity is

$$v_s = C_{ns}^T v_n. \tag{21}$$

The corrected vehicle-frame velocity is

$$v_c = C_{cs} v_s. \tag{22}$$

The optional wheel/CAN speed measurement observes

$$h_x(x) = e_x^T v_c. \tag{23}$$

The non-holonomic constraints use zero lateral and vertical vehicle velocity:

$$h_y(x) = e_y^T v_c \approx 0, \qquad h_z(x) = e_z^T v_c \approx 0. \tag{24}$$

For a left residual-mount perturbation,

$$q_{cs}^{true} = \delta q(\delta\psi_{cs}) \otimes q_{cs}, \tag{25}$$

the vehicle velocity perturbation is

$$v_c^{true} \simeq (I + [\delta\psi_{cs}]_\times) v_c = v_c - [v_c]_\times \, \delta\psi_{cs}. \tag{26}$$

Therefore the residual-mount Jacobian block for component i is

$$H_{\psi,i} = -e_i^T \, [v_c]_\times \, . \tag{27}$$

Runtime controls can ramp, gate, attenuate, or freeze the residual-mount part of the Kalman gain and injected correction for these rows.

6.4 Stationary Gravity

Stationary gravity updates use body/seed-frame accelerometer samples and update only x and y accelerometer channels:

$$f_s \approx -C_{ns}^T g_n + b_a + \eta. \tag{28}$$

The scalar rows are

$$h_x = e_x^T(-C_{ns}^T g_n + b_a), \qquad h_y = e_y^T(-C_{ns}^T g_n + b_a). \tag{29}$$

Before each stationary gravity scalar update, the runtime floors roll and pitch covariance to avoid over-confident attitude locking. The residual-mount Jacobian block is zero because the accelerometer sample has already been put in the seeded frame.

7 Injection and Covariance Reset

After each update the error state is injected:

$$q_{ns}^+ = q_{ns}^- \otimes \delta q(\delta\theta_s), \tag{30}$$

$$v_n^+ = v_n^- + \delta v_n, \qquad p_n^+ = p_n^- + \delta p_n, \tag{31}$$

$$b_g^+ = b_g^- + \delta b_g, \qquad b_a^+ = b_a^- + \delta b_a, \tag{32}$$

$$q_{cs}^+ = \delta q(\delta\psi_{cs}) \otimes q_{cs}^-. \tag{33}$$

Both quaternions are normalized. The attitude and residual-mount covariance blocks are reset with

$$G_r(\delta\theta) \simeq I - \frac{1}{2}\left[\delta\theta\right]_\times. \tag{34}$$

The full reset matrix is block diagonal with G_r on the attitude and mount blocks and identity elsewhere:

$$P^+ \leftarrow G_{\text{reset}} P^+ G_{\text{reset}}^T. \tag{35}$$

8 Initialization and Mount-Source Modes

At GNSS initialization the filter sets q_{ns} from GNSS heading or horizontal velocity course, copies GNSS NED position and velocity into p_n, v_n, sets biases to zero, and initializes $q_{cs} = 1$. Covariance is diagonal with configuration-controlled bias and residual-mount variances.

The fusion facade controls the coarse mount used for IMU pre-rotation:

- **Latched/internal**: align produces q_{vb}^0, which is frozen when the ESKF initializes. Residual mount q_{cs} may then estimate remaining seed error.
- **FollowAlign**: the pre-rotation source tracks align and the residual q_{cs} is repeatedly reset to identity. This is useful when the ESKF should not own mount estimation.
- **External/reference**: a fixed supplied mount or reference mount is used as the pre-rotation source.

9 Known Modeling Boundaries

- Residual mount does not affect inertial propagation; it is observable only through vehicle-frame velocity measurements and through cross-covariances.
- GNSS position, GNSS velocity, zero velocity, and stationary gravity do not directly observe q_{cs}.
- Lateral and vertical NHC observations can couple attitude, velocity, biases, and residual mount; aggressive mount updates are therefore gated and ramped in the runtime facade.
- The physical full mount used for plotting must account for the frozen coarse seed and the direction of q_{cs}. Under the convention above, a vehicle-to-body comparison uses the seed composed with the inverse residual, not the residual alone.